

Terminal Altitude Architecture and Post-Selection Artifacts in Adjacent Zeta-Zero Spacing

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Abstract

We re-examine extreme adjacent-spacing compression packets among Riemann zeta zeros in the terminal LMFDB band using a seam-stitched board of 935,111 rows spanning approximately 28.499547B to 30.610043B. Earlier exploratory projections suggested two positive structures: a narrow first-logarithmic-harmonic h_1 corridor and a discrete row-sector structure centered on modulus 1009. Subsequent search-adjusted validation audits show that neither interpretation survives as an explanatory mechanism. The circular resultant R is reproduced or exceeded by packet-count-matched search-adjusted synthetic and permutation controls; the apparent h_1 -specific Arc80 corridor is absorbed by rank-altitude and altitude-matched controls; and the $M = 1009$ sector signal fails a dense $M = 800, \dots, 1200$ search-adjusted permutation null.

The surviving empirical object is a multi-component terminal altitude architecture in the yield rate of highly compressed adjacent-spacing packets. A primary reconstructed packet population contains 4,676 extreme compression packets. Its altitude distribution is strongly non-uniform after normalization by board-row density, with descriptive Gaussian-mixture summaries favoring a multi-component altitude model over a single-component model. Altitude-only surrogate packet sets reproduce the previously observed h_1 Arc80 corridor, showing that h_1 is a descriptive projection of altitude localization rather than a coordinate-specific phase resonance. The resulting claim is finite-horizon and deliberately narrower: extreme compression packets form a terminal altitude architecture, while the earlier h_1 and modulus-1009 structures are examples of visually compelling post-selection artifacts in high-dimensional exploratory analysis of zeta-zero spacings.

Keywords: Riemann zeta zeros; adjacent zero spacing; extreme compression; terminal altitude architecture; post-selection inference; look-elsewhere effect; seam stitching; random matrix statistics; empirical number theory

1 Introduction

The local statistics of the nontrivial zeros of the Riemann zeta function have long been compared with random matrix ensembles, especially the Gaussian Unitary Ensemble (GUE) after unfolding [1, 2, 3]. These comparisons primarily concern bulk spacing statistics and correlation laws. The present work concerns a narrower empirical object: rare extreme compression events in adjacent zero spacing on a finite observed horizon.

An earlier version of this study reported that extreme compression packets appeared to organize in the first logarithmic harmonic

$$h_1(T) = \left\{ \frac{\log(T/2\pi)}{\pi} \right\}.$$

A continuation study then reported a possible row-sector structure centered on modulus 1009. Both interpretations were plausible exploratory descriptions, and both were supported by local nulls conditional on the chosen coordinate or modulus family. A blind-review critique correctly identified the central remaining weakness: the nulls did not fully simulate the act of searching over coordinate families, harmonic families, moduli, window sizes, and tail thresholds.

This replacement manuscript reports the result of the subsequent search-adjusted validation campaign. The original h_1 -specific interpretation does not survive. The $M = 1009$ row-sector interpretation does not survive. The underlying extreme-compression packet field, however, does survive. The data now support a different claim: the packets form a multi-component altitude architecture in the terminal band, and the old h_1 corridor is explained as a projection of that altitude architecture through a slowly varying logarithmic coordinate.

The paper therefore has two goals. First, it reports the surviving empirical structure: a terminal altitude architecture in the yield rate of highly compressed adjacent-spacing packets. Second, it documents the failure modes of two attractive but ultimately non-surviving projections: the continuous h_1 phase projection and the discrete modulus-1009 sector projection. The result is both an empirical finding and a methodological caution about post-selection inference in computational number theory.

1.1 What this manuscript replaces

The original preprint framed the terminal packet field as evidence for a coordinate-specific h_1 phase-locking phenomenon. That language is no longer maintained. In the present manuscript:

- the h_1 coordinate is retained only as a descriptive projection;
- the circular resultant R is demoted to a diagnostic statistic;
- the narrow h_1 Arc80 corridor is explained by altitude localization;
- the modulus-1009 row-sector result is rejected under dense search-adjusted controls;
- the surviving positive claim is terminal altitude architecture of extreme compression packets.

1.2 Scope

All claims are finite-horizon empirical claims on the loaded terminal board. No asymptotic theorem is claimed. No proof or disproof of the Riemann Hypothesis is claimed. The work assumes the tabulated zeros used in the computation are zeros on the critical line and studies their adjacent spacings as an empirical sequence.

2 Data and seam-stitched board

2.1 Source horizon

The terminal board used in the validation campaign contains 935,111 rows and spans approximately

$$T = 28.499547B \quad \text{to} \quad T = 30.610043B.$$

The data are drawn from the available LMFDB zero tables, supplemented conceptually by the classical Odlyzko zero-table framework [5, 6]. The working filenames used during the computational campaign sometimes contain the string **31p5B**; this was a downloader ceiling, not a claim that source data extend to 31.5B. The actual terminal ordinate in the loaded source sequence is approximately 30.610B.

2.2 Seam stitching

The board is seam stitched: consecutive zero ordinates are decoded in shard order and cross-shard adjacent gaps are retained. This matters because discarding cross-shard gaps would create artificial seams in any local spacing statistic. The seam-preserving design is therefore part of the empirical instrument, not a cosmetic preprocessing step.

2.3 Terminal row metric

The terminal packet audits use row-level spacing ratios derived from fitted and pure spacing spines. The primary metric in the final validation chain is `Gap_to_Fitted_Ratio`; `Gap_to_Pure_Ratio` is retained as a diagnostic comparator. Extreme packets are extracted by thresholding the row-level score distribution at high quantiles and grouping adjacent selected rows into connected packets.

The primary row-level reconstruction used in the validation chain selects the upper tail of `Gap_to_Fitted_Ratio` at

$$\tau = 0.995,$$

yielding 4,676 packets. This is close to the original terminal packet count used in the earlier phase-locking analysis, and it avoids the one-packet degeneracy encountered by an initial rolling-window proxy benchmark.

3 Methods

3.1 Packet extraction

For a row-level score sequence X_k , threshold τ , and tail direction d , define the selected row set

$$\mathcal{E}_{\tau,d} = \begin{cases} \{k : X_k \geq q_\tau(X)\}, & d = \text{ge}, \\ \{k : X_k \leq q_{1-\tau}(X)\}, & d = \text{le}. \end{cases}$$

A packet is a connected component of $\mathcal{E}_{\tau,d}$ in row-index order. The packet center is chosen as the most extreme row within the component.

The primary population in this manuscript uses `Gap_to_Fitted_Ratio`, $d = \text{ge}$, and $\tau = 0.995$. Sensitivity grids over nearby τ values and over the pure-ratio score are used to distinguish stable architecture from metric-specific endpoint behavior.

3.2 Circular statistics

For packet altitudes $T_1, \dots, T_N$, the earlier preprint studied the phases

$$h_m(T_j) = \left\{ m \frac{\log(T_j/2\pi)}{\pi} \right\}.$$

The circular resultant is

$$R = \left| \frac{1}{N} \sum_{j=1}^N e^{2\pi i h_m(T_j)} \right|.$$

For $q = 0.80$, Arc80 is the shortest circular arc covering 80% of the packet phases.

The validation campaign shows that R is not a robust positive statistic under search-adjusted nulls. Arc80 remains useful, but the h_1 Arc80 corridor is explained by altitude localization rather than by h_1 -specific resonance.

3.3 Packet yield rate

A central correction in this replacement manuscript is the use of packet yield rate rather than raw packet count. For an altitude bin B , define

$$Y(B) = \frac{\#\{\text{packet centers in } B\}}{\#\{\text{board rows in } B\}}.$$

This avoids the denominator trap: zero density varies with altitude, so raw packet counts alone cannot distinguish true packet overrepresentation from a larger row population.

3.4 Altitude-only projection null

To test whether the h_1 Arc80 corridor is explained by altitude localization, we generate surrogate packet sets that preserve the packet altitude histogram. The terminal altitude range is divided

into bins; within each bin, the surrogate draws the same number of rows as the real packet field contributes from that bin. The surrogate packet altitudes are then mapped through h_1 , and Arc80 is recomputed.

If these altitude-only surrogates reproduce the real h_1 Arc80, then the h_1 corridor is explained as a projection of altitude localization.

3.5 Search-adjusted harmonic null

The h_1 claim is vulnerable to a look-elsewhere effect over harmonics. The validation campaign therefore allows synthetic and permutation nulls to scan

$$h_1, h_2, h_3, h_4, h_5$$

and retain the best circular statistic. This compares the real result to the null’s best available harmonic, not merely to the null’s h_1 value.

3.6 Coordinate-specificity audit

To test whether the Arc80 corridor belongs specifically to h_1 , the validation campaign compares h_1 against:

- nearby frequency perturbations $h_{1,\epsilon}(T) = \{(1 + \epsilon) \log(T/2\pi)/\pi\}$;
- higher harmonics h_m ;
- monotone altitude controls, including linear T , $\sqrt{T}$, $\log T$, and rank- T ;
- random phase controls.

The key normalized statistic is

$$\text{Arc80}_{\text{norm}} = \frac{\text{Arc80}(\text{packet phases})}{\text{Arc80}(\text{all board rows})}.$$

This prevents coordinates with small total phase motion over the terminal band from appearing artificially strong.

3.7 Dense modulus search-adjusted null

For the row-sector claim, the validation campaign scans all moduli

$$M = 800, 801, \dots, 1200$$

under a base-6 sectorization and computes $S_{\text{sector}}(M)$. Each permutation null shuffles tail signs, scans the same full modulus range, and records

$$\max_{800 \leq M \leq 1200} S_{\text{sector}}^{\text{null}}(M).$$

This is the correct look-elsewhere comparison for the modulus claim.

4 Validation chronology

Table 1 summarizes the validation sequence.

Audit	Question	Result	Role
P83B	Does h_1 concentration survive packet-count-matched search-adjusted synthetic controls?	Split result: R fails; Arc80 survives packet-count controls.	Demotes R ; isolates Arc80 for further testing.
P84	Is the surviving Arc80 corridor h_1 -specific?	No. Rank/altitude controls and block-matched nulls absorb h_1 specificity.	Rejects coordinate-specific h_1 interpretation.
P85	What explains the h_1 corridor if h_1 is not the mechanism?	Altitude-only surrogates reproduce h_1 Arc80; packet field has multi-component altitude architecture.	Establishes replacement mechanism.
P85R	Is there extra sub-block microstructure beyond macro altitude bands?	No strong support after repaired sub-block null.	Keeps replacement mechanism macro-altitude, not micro-fractal.
P86	Does modulus 1009 survive dense search-adjusted modulus controls?	No. No modulus survives look-elsewhere correction.	Rejects row-sector engine claim.

Table 1: Validation chronology leading to the replacement interpretation.

5 Results

5.1 The packet field survives

The first key result is that the packet object survives the interpretive collapse of the h_1 coordinate. The P83B reconstruction extracts 4,676 packets under the primary fitted-ratio tail rule. This establishes that the empirical object is not merely an artifact of a rolling-window proxy or a one-packet degeneracy.

Quantity	Value
Terminal rows	935,111
Altitude range	28.499547B to 30.610043B
Primary metric	Gap_to_Fitted_Ratio
Primary direction	ge
Primary threshold	$\tau = 0.995$
Primary packet count	4,676

Table 2: Primary terminal packet reconstruction used in the validation chain.

5.2 R fails, Arc80 survives packet-count controls

P83B compares the packet field against packet-count-matched synthetic and permutation controls, with the null allowed to scan harmonics $h_1, \dots, h_5$. The result is split. The circular resultant R does not separate from the search-adjusted null. In fact, the null routinely matches or exceeds the real resultant. The Arc80 corridor, however, is narrower than the packet-count-matched synthetic and permutation controls.

This split changes the role of the statistics. R is retained only as descriptive. Arc80 remains a real geometric feature, but it requires coordinate-specificity testing.

5.3 The Arc80 corridor is not h_1 -specific

P84 tests whether the surviving Arc80 corridor belongs specifically to h_1 . It does not. The normalized h_1 Arc80 is outperformed by rank-altitude controls, and block-matched altitude nulls absorb the apparent h_1 -specific effect. The epsilon perturbation surface is effectively flat, indicating that h_1 is not a sharp local optimum in the nearby log-phase family.

The conclusion is not that the packet field is random. It is that h_1 is not the explanatory coordinate.

5.4 Altitude-only surrogates reproduce the h_1 corridor

P85 provides the replacement explanation. The real h_1 Arc80 for the primary packet field is approximately

$$\text{Arc80}_{h_1} = 0.017454.$$

Altitude-only surrogate packets, drawn to preserve the same altitude-bin distribution, have mean

$$\text{Arc80}_{h_1}^{\text{sur}} \approx 0.017469.$$

The empirical probability that the surrogate Arc80 is at least as tight as the real value is

$$p \approx 0.1417165669.$$

Thus the old h_1 corridor is reproduced by altitude localization alone. The h_1 coordinate is a descriptive projection of the packet altitude distribution.

The numerical proximity

$$0.1417165669 \approx \pi - 3$$

is recorded as a future-hypothesis marker only. It is not used as evidence in the present statistical interpretation.

Projection statistic	Value
Real h_1 Arc80	0.017454
Altitude-only surrogate mean h_1 Arc80	0.017469
Empirical probability surrogate $\leq$ real	0.1417165669
$\pi - 3$ reference	0.1415926536
Interpretation	h_1 Arc80 explained by altitude projection

Table 3: Projection explanation of the old h_1 Arc80 corridor. The surrogate preserves packet altitude distribution but randomizes row choice within altitude bins.

5.5 Altitude architecture of the packet field

The packet yield rate is not uniform across altitude. P85 computes

$$Y(B) = \frac{\#\{\text{packets in altitude bin } B\}}{\#\{\text{board rows in altitude bin } B\}},$$

avoiding raw-count bias from changing row density. The yield-rate maps show broad altitude bands of elevated and suppressed packet frequency.

A Gaussian-mixture descriptive summary favors a multi-component model. In the primary run, $K = 4$ is selected by BIC, with a BIC improvement over $K = 1$ of approximately

$$2823.78.$$

This is not interpreted as a theorem that exactly four physical bands exist. It is interpreted as strong descriptive evidence that a single unimodal altitude model is inadequate.

Altitude-architecture statistic	Value
Primary packet count	4,676
Best descriptive GMM K	4
BIC improvement $K = 1$ minus best K	2823.78
Sub-block microstructure support after macro-block matching	Not supported
Interpretation	Macro-scale terminal altitude architecture

Table 4: Descriptive altitude-architecture summary.

5.6 No strong evidence for extra sub-block micro-clustering

The original P85 block-null statistic was degenerate because the null preserved the same block counts used by the test statistic. P85R repaired this by matching macro-block counts while testing finer micro-bin and nearest-neighbor statistics. The repaired test does not find strong evidence for extra fine-scale clustering beyond the macro altitude architecture.

Representative repaired sub-block p -values are:

$$p_{\text{Gini}} \approx 0.180, \quad p_{\text{max}} \approx 0.315, \quad p_{\text{PNN,mean}} \approx 0.244, \quad p_{\text{PNN,median}} \approx 0.353.$$

This simplifies the replacement interpretation: the surviving object is macro-altitude architecture, not confirmed fractal or micro-scale clustering.

5.7 Tail-polarity altitude asymmetry

The primary fitted-ratio upper and lower tails are not identical in altitude distribution. In the primary $\tau = 0.995$ fitted-ratio comparison, the two tails have a Kolmogorov–Smirnov-type distance of approximately

$$0.168$$

and a Jensen–Shannon divergence of approximately

$$0.0496.$$

This supports a descriptive tail-polarity asymmetry in altitude. The effect is not stated as a causal thermodynamic mechanism. It is a finite-horizon distributional asymmetry between the two fitted-ratio tails.

5.8 The modulus-1009 row-sector claim fails

P86 scans all moduli $M = 800, \dots, 1200$. Each permutation null is allowed to scan the same modulus range and record its maximum sector amplitude. Under this search-adjusted null, neither $M = 1009$ nor any real best modulus survives.

Across the four seam-scale windows $W = 900, 1000, 1100, 1300$, the number of configurations in which $M = 1009$ survives the search-adjusted null is

$$0.$$

The number of configurations in which the best real modulus survives the search-adjusted null is also

$$0.$$

Thus the row-sector continuation paper’s positive $M = 1009$ interpretation is rejected as a search-hardened discovery.

Window	Best real modulus	Rank of 1009	p_{best}	p_{1009}
900	1133	48	0.0798	1.0000
1000	1087	151	0.2355	1.0000
1100	1133	143	1.0000	1.0000
1300	1022	58	0.9381	1.0000

Table 5: Dense modulus validation summary. The search-adjusted null scans $M = 800, \dots, 1200$ on every permutation. Neither $M = 1009$ nor the best real modulus survives.

6 Claim transition ledger

Table 6 summarizes the shift from the earlier preprint interpretation to the present replacement interpretation.

Old claim	Validation result	Revised claim
Extreme compression packets phase-lock to h_1 .	P83B and P84 reject R and h_1 -specificity under search-adjusted and coordinate-control nulls.	Rejected as a coordinate-specific phase-locking resonance.
The h_1 Arc80 corridor is intrinsically h_1 -specific.	P85/P85R show that altitude-only surrogates reproduce the h_1 Arc80 corridor.	h_1 Arc80 is a descriptive projection of altitude-localized packet architecture.
The packet object disappears if h_1 fails.	P85 shows packet yield-rate maps and multi-component altitude structure.	The core object survives as terminal altitude architecture of highly compressed adjacent-spacing packets.
$M = 1009$ is a row-sector engine.	P86 dense modulus scan finds no search-adjusted survival.	Rejected as a look-elsewhere artifact in sparse tail-sector scanning.

Table 6: Claim transition ledger from the earlier preprint interpretation to the present replacement interpretation.

7 Discussion

The validation campaign changes the interpretation of the original discovery but does not erase the empirical object. The earlier h_1 analysis detected a real pattern because the packets were not uniformly distributed in altitude. The mistake was to assign the explanatory role to h_1 itself. The revised interpretation is more conservative: h_1 is a projection through which altitude localization became visible.

The modulus-1009 claim fails more completely. Once the null is allowed to search the same dense modulus range, the apparent sector engine disappears into the look-elsewhere envelope. Unlike the h_1 corridor, which is explained as a projection of a surviving altitude architecture, the modulus-1009 interpretation does not currently have a surviving positive replacement. It remains valuable as a methodological warning.

The central surviving result is therefore the altitude architecture of extreme compression packets. This is a finite-horizon empirical structure. It may reflect nonstationarity in the terminal segment, long-range fluctuation in the observed zero sequence, or a more subtle interaction between local spacing tails and altitude. The present paper does not identify a theoretical mechanism for the altitude bands. It establishes that the bands exist in the loaded terminal board and that they explain the earlier h_1 projection.

7.1 Relation to GUE statistics

The results do not claim a breakdown of GUE bulk statistics. The packet field lies in the rare-event tail of adjacent spacing behavior. The observation is that the extreme compression packets are not uniformly distributed over the terminal board. This is compatible with bulk GUE-like behavior while still allowing finite-horizon tail architecture.

7.2 Lessons for empirical number theory

This study demonstrates a general methodological point. Conditional nulls can be misleading after a large exploratory search. A null that tests a chosen coordinate after the coordinate has already been selected does not address the look-elsewhere effect. For empirical studies of zeta zeros, phase coordinates, modulus scans, and geometric projections should be evaluated using nulls that are allowed to search the same family of alternatives.

8 Limitations

The analysis is finite-horizon. The terminal board ends near 30.610B, and no claim is made beyond the loaded source range. The altitude architecture may change, vanish, or sharpen at greater heights.

The packet extractor is empirical. Although it has been stress-tested across nearby thresholds and metrics, it is not derived from a theorem about zeta zeros. The GMM component count is descriptive, not a proof of exactly four physical bands.

The h_1 projection explanation is strong on the observed board, but it does not imply that all future h_1 -like behavior is an artifact. It means the h_1 behavior measured in the present terminal packet field is explained by altitude localization.

The modulus-1009 failure is specific to the tested row-sector pipeline and dense modulus search. It rejects the positive claim made by the continuation paper but does not rule out all possible arithmetic sectorizations under different, pre-specified theoretical frameworks.

9 Conclusion

The original h_1 phase-locking interpretation and the $M = 1009$ row-sector interpretation do not survive search-adjusted validation. The surviving structure is a terminal altitude architecture in highly compressed adjacent-spacing packets. The packet field is non-uniform over altitude, has multi-component descriptive structure, and explains the previously observed h_1 Arc80 corridor through altitude projection.

The revised conclusion is therefore narrower but stronger: the terminal band contains a finite-horizon architecture of extreme compression packets, and this architecture can cast misleading shadows in both continuous logarithmic phase coordinates and discrete modulus coordinates. Search-adjusted validation separates the surviving object from its projections.

A Audit glossary

Audit	Role
P83B	Packet-count-matched search-adjusted harmonic benchmark. Demotes R , preserves Arc80 as a statistic requiring coordinate testing.
P84	Coordinate-specificity audit. Rejects h_1 -specific explanation of the Arc80 corridor.
P85	Altitude architecture audit. Establishes packet yield-rate altitude structure and shows altitude-only surrogates reproduce h_1 Arc80.
P85R	Cleanup audit. Repairs the degenerate block-null and produces the claim transition ledger.
P86	Dense modulus search-adjusted null. Rejects $M = 1009$ as a search-hardened sector-engine discovery.
P87	Packaging scaffold. Consolidates the surviving and rejected claims into this replacement manuscript structure.

B Notation

Symbol	Meaning
γ_n	ordinate of the n -th nontrivial zero on the critical line
g_n	adjacent-zero gap $\gamma_{n+1} - \gamma_n$
T	representative altitude
$h_m(T)$	logarithmic harmonic phase coordinate
R	circular resultant
Arc80	shortest circular arc covering 80% of packet phases
$Y(B)$	packet yield rate in altitude bin B
τ	row-level tail threshold quantile

Table 8: Principal notation.

Data and Artifact Availability

The pre-validation preprint was deposited under DOI [10.5281/zenodo.19827222](https://doi.org/10.5281/zenodo.19827222). The validation campaign generates the replacement claim ledger and manuscript scaffold in the ForgeV16c audit chain, including outputs from P83B, P84, P85, P85R, P86, and P87. The source zero ordinate data are attributed to LMFDB and Odlyzko zero tables [5, 6].

Use of AI Tools

Generative AI tools were used for drafting assistance, code critique, adversarial claim review, and organization of the validation ledger. All research design decisions, computational execution choices,

interpretation, and final manuscript responsibility remain with Jeffery Huckstead.

Author Contributions

Jeffery Huckstead conceived the research program, assembled and audited the seam-stitched board, executed the computational validation campaign, interpreted the results, and wrote the manuscript.

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